

# How STARTUPS Can HASTEN DESIGN WITHOUT Burning Through FUNDS



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Developing a project from scratch is every designer's dream. Unfortunately, most ideas are not designed for manufacturing from the beginning. Often, the changes required are significant enough to prevent a product from even shipping. Tony Fadell, known as one of fathers of iPod, said at a conference, "There is a reason they call it hardware—it is hard."

This does not mean it is impossible, but the designers require access to not only the components and equipment for design, but also the know-how to put it together into a functional product. This means ensuring that their design can be built by the limited capabilities of a manufacturing facility or withstand the noisy analogue world we live in.

This article will give you an industry veteran's take on the best way to reduce time-to-market for a successful design while ensuring you do not draw flak from your investors.

## OpEx is better than CapEx

While there are advantages to having state-of-the-art test equipment in your

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lab, design houses still find it difficult to get experienced design and test engineers who are capable of crafting test strategies and executing test automation and design. Adding the cost of experienced engineers and high-end equipment together is enough to make most angel investors fly away.

One way to overcome this challenge is to outsource certain elements of design and test to service providers. This ensures that your team focuses on ensuring that the core competencies of your product are delivered.

## Everyone loves a smaller bill

There are two ways in which you can lower costs. One is by having your bill of materials (BoM) optimised, and second is by reducing the design cycle time so that you are able to push out two projects or revisions in the time of one.

Most startups begin prototyping on Arduino or use 3D printing for certain components. These methods are almost always way too expensive to be taken to market and do not scale well. You need to move on to a custom-designed circuit board that delivers the feature set you need and nothing else. This is where elements like the power supply and peripherals that you took for granted earlier begin hatching plans to ditch you.

What you could do here is to put together the proof-of-concept, no matter how bulky it is, and then approach the manufacturer to communicate your idea. The manufacturer will help you in refin-

